

Proof Writing

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This page collects general advice on how to write [proofs](#) and [derivations](#). The goal of a proof is not just to convince yourself that a [result](#) is true; it is to convince a *reader*, and to show them *why* it is true. Each principle on this page serves that goal. The symbols for logical entailment are listed under [Proofs](#) on the Notation page.

Don't skip steps

Show every step of the derivation. Each line should follow from the previous line by a single, identifiable move:

- applying a definition,
- substituting a known result, or
- performing one algebraic operation.

When you skip steps, you force the reader to reconstruct your reasoning, which is exactly the work the proof was supposed to do *for* them. A step that feels “obvious” while you are writing it is often the step a reader gets stuck on. Phrases like “it can be shown that” or “clearly” usually mark a skipped step; show the step instead.

Annotate each step

For each step, note the definition, [theorem](#), or algebraic rule that justifies it. A brief annotation to the right of the line is usually enough. Annotations make the proof checkable, and they teach the reader which tool to reach for in similar situations.

For example, here is a derivation that the [hat matrix](#) $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ is idempotent, $\mathbf{H}^2 = \mathbf{H}$, with each step annotated:

$$\begin{aligned}
\mathbf{H}^2 &= \mathbf{H}\mathbf{H} && \text{(definition of a matrix power)} \\
&= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(definition of } \mathbf{H} \text{)} \\
&= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{X}^\top \mathbf{X}) (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(regroup; matrix multiplication is associative)} \\
&= \mathbf{X} \mathbf{I}_p (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(a matrix inverse times the matrix is } \mathbf{I}_p \text{)} \\
&= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(the identity matrix changes nothing)} \\
&= \mathbf{H} && \text{(definition of } \mathbf{H} \text{)}
\end{aligned}$$

The [Linear Algebra](#) page uses this derivation to show that $\mathbf{H}$ is a projection matrix. The probability notes derive the variance identity $\text{Var}(X) = \text{E}[X^2] - (\text{E}[X])^2$ [in the same annotated style](#).

Proof by induction

Definition 0.1 (Proof by induction). To prove that a statement $P(n)$ holds for every integer $n \geq n_0$, a **proof by induction** on n proves two things:

1. the **base case**: $P(n_0)$ holds;
2. the **inductive step**: for every $n \geq n_0$, if $P(n)$ holds, then $P(n+1)$ holds. The assumption that $P(n)$ holds is called the **induction hypothesis**.

Together these give $P(n)$ for every $n \geq n_0$: the base case gives $P(n_0)$, the inductive step then gives $P(n_0+1)$, then $P(n_0+2)$, and so on.

Example 0.1 (The sum of the first n positive integers). Claim: for every integer $n \geq 1$,

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

Base case ($n = 1$): the left side is 1, and the right side is $\frac{1 \cdot 2}{2} = 1$.

Inductive step: assume the claim holds for n . Then

$$\begin{aligned}
\sum_{i=1}^{n+1} i &= \sum_{i=1}^n i + (n+1) && \text{(split off the last term)} \\
&= \frac{n(n+1)}{2} + (n+1) && \text{(induction hypothesis)} \\
&= \frac{n(n+1)}{2} + \frac{2(n+1)}{2} && \text{(write } n+1 \text{ over the denominator 2)} \\
&= \frac{n(n+1) + 2(n+1)}{2} && \text{(add fractions with the same denominator)} \\
&= \frac{(n+2)(n+1)}{2} && \text{(factor out } n+1 \text{)} \\
&= \frac{(n+1)(n+2)}{2} && \text{(reorder the factors)} \\
&= \frac{(n+1)((n+1)+1)}{2} && \text{(write } n+2 \text{ as } (n+1)+1 \text{)}
\end{aligned}$$

which is the claim for $n+1$.

Follow the golden rule

In general, follow the golden rule: treat your readers the way you want to be treated as a reader.

When you read someone else's proof, you want to be able to follow every step without guessing, to know which result is being used at each line, and to never be left wondering where a quantity came from. Write your own proofs to meet that same standard.

Proof by induction

Many proofs in these notes show that a statement holds for every **natural number** n , or for every integer n from some starting value on. Checking the statement one n at a time would never finish; induction proves it for all of them with two arguments.

Definition 0.2 (Proof by induction). Let n_0 be an integer, and for each integer $n \geq n_0$ let $P(n)$ be a statement about n . A **proof by induction** on n shows that $P(n)$ holds for every integer $n \geq n_0$ by showing two things:

1. **Base case:** $P(n_0)$ holds.
2. **Inductive step:** for every integer $n > n_0$, if $P(n-1)$ holds, then $P(n)$ holds.

In the inductive step, the assumption that $P(n-1)$ holds is called the **induction hypothesis**.

Remark 0.1 (Why the two steps suffice). The base case gives $P(n_0)$. The inductive step with $n = n_0 + 1$ then gives $P(n_0 + 1)$, with $n = n_0 + 2$ it gives $P(n_0 + 2)$, and so on: each integer $n \geq n_0$ is reached after $n - n_0$ uses of the inductive step.

When $P(n)$ is needed only for $n_0 \leq n \leq r$, it is enough to show the inductive step for $n_0 < n \leq r$; the same chain then stops at $P(r)$.

The following example uses [summation notation](#).

Example 0.2 (Sum of the first n natural numbers). For every $n \in \mathbb{N}$,

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}.$$

For example, with $n = 3$, $1 + 2 + 3 = 6$ and $\frac{3 \cdot 4}{2} = 6$. To prove the equation for every n , let $P(n)$ be the equation and use Definition 0.2 with $n_0 = 1$.

Base case, $n = 1$. The sum has one term:

$$\sum_{i=1}^1 i = 1 \quad (\text{definition of a sum})$$

and the right side is

$$\begin{aligned} \frac{1(1+1)}{2} &= \frac{1 \cdot 2}{2} & (1+1=2) \\ &= \frac{2}{2} & (1 \cdot 2 = 2) \\ &= 1. & (\frac{2}{2} = 1) \end{aligned}$$

So $P(1)$ holds.

Inductive step. Let $n \geq 2$, and suppose $P(n-1)$ holds:

$$\sum_{i=1}^{n-1} i = \frac{(n-1)((n-1)+1)}{2}.$$

Then

$$\begin{aligned}
\sum_{i=1}^n i &= 1 + 2 + \cdots + (n-1) + n && \text{(definition of a sum)} \\
&= (1 + 2 + \cdots + (n-1)) + n && \text{(associative law)} \\
&= \sum_{i=1}^{n-1} i + n && \text{(definition of a sum)} \\
&= \frac{(n-1)((n-1)+1)}{2} + n && \text{(induction hypothesis)} \\
&= \frac{(n-1)n}{2} + n && ((n-1)+1 = n) \\
&= \frac{(n-1)n}{2} + \frac{2n}{2} && (n = \frac{2n}{2}) \\
&= \frac{(n-1)n + 2n}{2} && \text{(common denominator)} \\
&= \frac{((n-1)+2)n}{2} && \text{(distributive law)} \\
&= \frac{(n+1)n}{2} && ((n-1)+2 = n+1) \\
&= \frac{n(n+1)}{2}, && \text{(commutative law)}
\end{aligned}$$

which is $P(n)$. By Definition 0.2, the equation holds for every $n \in \mathbb{N}$.

Further reading

- Velleman (2019) is a structured introduction to proof techniques such as direct proof, [proof by contradiction](#), and induction.
- Barker-Plummer et al. (2011) treats proofs as formal derivations in propositional and first-order logic.

References

- Barker-Plummer, Dave, Jon Barwise, and John Etchemendy. 2011. *Language, Proof and Logic*. 2nd ed. CSLI Publications. <https://www.amazon.com/dp/1575866323>.
- Velleman, Daniel J. 2019. *How to Prove It: A Structured Approach*. 3rd ed. Cambridge University Press. <https://doi.org/10.1017/9781108539890>.